

AWS Basics

A Beginner's Learning Guide to Amazon Web Services

Covers: cloud concepts, global infrastructure, IAM, EC2, S3, VPC, RDS, Lambda, CloudWatch, billing safety, and a first hands-on lab.

1. What is Cloud Computing and AWS?

Cloud computing means renting computing resources (servers, storage, databases, networking) over the internet and paying only for what you use, instead of buying and maintaining physical hardware.

Amazon Web Services (AWS) is the largest cloud provider, offering 200+ services.

Key benefits

- **Pay-as-you-go:** no big upfront cost.
- **Scalable:** grow or shrink resources in minutes.
- **Global:** deploy close to users worldwide.
- **Reliable and secure:** built-in redundancy and security tooling.

Service models

Model	You manage	Example
IaaS	OS, apps, data	EC2 (virtual servers)
PaaS	Apps and data	Elastic Beanstalk, RDS
SaaS	Just use it	WorkMail, Amazon Chime

Shared Responsibility Model

AWS is responsible for security **OF** the cloud (data centers, hardware, core network). You are responsible for security **IN** the cloud (your data, IAM users, passwords, firewall rules, OS patches on EC2).

2. Global Infrastructure

- **Region:** a geographic area with multiple data centers (e.g. Mumbai - ap-south-1).
- **Availability Zone (AZ):** one or more isolated data centers inside a Region. Spread resources across AZs for high availability.
- **Edge Locations:** used by CloudFront (CDN) to cache content close to users.

Tip: always check which Region is selected (top-right of the console) before creating anything. Resources in other Regions will not show up in your current view.

3. Getting Started Safely

- Create an AWS account and enable **MFA** on the root user.
- Do **not** use the root user for daily work. Create an IAM user or role instead.
- Set a **billing budget and alert** (AWS Budgets) before launching anything.
- Check the current Free Tier terms on the AWS Free Tier page, as offers change over time.
- Always **stop or terminate** resources when finished (EC2, RDS, Elastic IPs, NAT Gateways can all keep costing money).

4. Core Services

4.1 IAM - Identity and Access Management

Controls **who** can do **what** in your account. IAM is global (not tied to a Region).

Concept	Meaning
User	A person or app with long-term credentials
Group	A collection of users sharing permissions
Role	Temporary permissions assumed by users or services (e.g. EC2 accessing S3)
Policy	A JSON document defining allowed or denied actions

Golden rule: **least privilege** - give only the permissions that are needed.

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": "s3:GetObject",
    "Resource": "arn:aws:s3:::my-bucket/*"
  }]
}
```

4.2 EC2 - Elastic Compute Cloud

Virtual servers in the cloud. Key pieces:

- **AMI:** template (OS plus software) used to launch an instance.
- **Instance type:** CPU/RAM size, e.g. t2.micro or t3.micro (small, often Free Tier eligible).
- **Key pair:** used to SSH into Linux instances.
- **Security Group:** a virtual firewall controlling inbound/outbound traffic (e.g. allow port 22 for SSH, 80 for HTTP).
- **EBS:** block storage volumes attached to instances.
- **Pricing:** On-Demand, Reserved/Savings Plans, Spot (cheap but can be interrupted).

4.3 S3 - Simple Storage Service

Object storage for files of any type. Data lives in **buckets** (globally unique names) as **objects** (file plus metadata). Highly durable and cheap.

- Uses: backups, static website hosting, data lakes, logs, media files.
- Storage classes: Standard, Intelligent-Tiering, Standard-IA, Glacier (archival).
- Buckets are private by default. Keep "Block Public Access" on unless you truly need a public site.
- Enable **versioning** to recover from accidental deletes.

4.4 VPC - Virtual Private Cloud

Your own isolated network inside AWS.

- **Subnet:** a slice of the VPC in one AZ (public or private).

- **Internet Gateway:** lets a public subnet reach the internet.
- **Route Table:** rules for where network traffic goes.
- **NAT Gateway:** lets private subnets reach out to the internet (costs money, so be careful).
- **Security Groups** (stateful, instance level) vs **NACLs** (stateless, subnet level).

4.5 RDS - Relational Database Service

Managed databases (MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Aurora). AWS handles backups, patching, and failover. Use **Multi-AZ** for high availability and **Read Replicas** for read scaling. Related: **DynamoDB** is a managed NoSQL key-value database.

4.6 Lambda - Serverless Compute

Run code without managing servers. You upload a function; AWS runs it in response to events (an S3 upload, an API call, a schedule) and bills per execution time.

```
import json

def lambda_handler(event, context):
    return {
        "statusCode": 200,
        "body": json.dumps("Hello from Lambda!")
    }
```

4.7 CloudWatch - Monitoring

Collects metrics, logs, and alarms. Example: alarm when EC2 CPU stays above 80 percent, or notify you when your estimated bill crosses a limit.

4.8 Other services worth knowing

Service	Purpose
Route 53	DNS and domain management
CloudFront	Content delivery network (CDN)
ELB	Load balancer spreading traffic across instances
Auto Scaling	Add or remove EC2 instances automatically
SNS / SQS	Notifications / message queues
CloudFormation	Infrastructure as Code (templates)
ECS / EKS	Run containers (Docker / Kubernetes)

5. AWS CLI Basics

The CLI lets you control AWS from a terminal.

```
aws configure                # enter access key, secret, region
aws sts get-caller-identity  # check who you are logged in as
aws s3 ls                    # list buckets
aws s3 mb s3://my-unique-bucket # make a bucket
aws s3 cp file.txt s3://my-unique-bucket/
aws ec2 describe-instances  # list EC2 instances
```

Prefer IAM users/roles with limited permissions for CLI keys. Never commit access keys to GitHub.

6. Hands-on Lab: Host a Static Website on S3

- Sign in to the console and choose a Region near you.
- Open **S3** and click **Create bucket** with a globally unique name.
- Upload an `index.html` file.
- In the bucket's **Properties** tab, enable **Static website hosting** and set `index.html` as the index document.
- Adjust Block Public Access and add a bucket policy that allows public read (learning only).
- Open the website endpoint URL to see your page.
- **Clean up:** empty and delete the bucket when finished.

Second lab idea: launch a free-tier-eligible EC2 instance, SSH in, install nginx, view the default page in the browser, then terminate the instance.

7. Cost and Security Checklist

- MFA on root; no daily use of root.
- Budget alert created.
- Least-privilege IAM permissions.
- S3 buckets private unless intentionally public.
- No access keys or passwords in code or GitHub repos.
- Terminate unused EC2, RDS, load balancers, NAT gateways, and Elastic IPs.
- Check the Billing dashboard regularly.

8. Suggested Learning Path

- Cloud concepts and Shared Responsibility Model
- IAM, then EC2, then S3
- VPC networking, then RDS
- Lambda, CloudWatch, then CloudFormation
- Prepare for **AWS Certified Cloud Practitioner (CLF-C02)**, then Solutions Architect Associate

Resources

- AWS Skill Builder (free digital training): skillbuilder.aws
- AWS Documentation: docs.aws.amazon.com
- AWS Well-Architected Framework: aws.amazon.com/architecture/well-architected
- AWS Free Tier: aws.amazon.com/free

These notes are a starting point. Always verify pricing and Free Tier limits on the official AWS pages, as they change.